

QXU6008 Materials and Sustainability

2025B | Prof. Maria Romero-Gonzalez and Prof. Giuseppe Viola

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1. Introduction to Module

■ Assessment

Activity	Score
Group Coursework	30%
-Peer Assessment	10%
Individual Coursework	40%
Exam in Week10	20%

2. Resource Consumptionb

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3. Business and Sustainability

New Tech Capabilities	=	New Ways of Consuming	=	New Business Models
New Social Agendas		New Ways of Producing		New Design Approaching

3.1. Linear Economy

Resource Extraction → Manufacture → Distribution & Retail → Consumer Use → Disposal

Advantages	Disadvantages
- Established and familiar economic model.	- High resource consumption leading to depletion.
- Straightforward production and consumption processes.	- Generates significant waste, impacting the environment.
- Relatively simple supply chains and infrastructure.	- Vulnerability to resource scarcity and price fluctuations.
	- Limited focus on long-term sustainability and resilience.
	- Contributes to environmental degradation and pollution.

3.2. Collaborative Economy

An economy built on distributed networks of connected individuals and communities versus centralized institutions, transforming how we can produce, consume, finance and learn.

■ Different funding models:

- **Reward-Based:** Funders receive non-financial incentives or rewards.
- **Equity-Based:** Funders receive equities or shares in the project or company.
- **Debt-Based (Crowdlending):** Funders lend money with repayment expectation with interest.
- **Donation-Based:** Funders do not expect financial returns.